

The Complex Leech Lattice Detects the Weight-12 Level-3 Newform

A fully verified computation

Abstract

We give a completely computational, independently cross-checked account of how the *complex Leech lattice*—the rank-12 $\mathbb{Z}[\omega]$ -structure carried by the Leech lattice through an order-3 fixed-point-free isometry—detects the unique cuspidal newform $g \in S_{12}(\Gamma_0(3))$. We build the Leech lattice Λ from the binary Golay code and certify it three independent ways: kissing number 196560, automorphism group $\text{Aut}(\Lambda) = \text{Co}_0$, and the full minimal-vector inner-product distribution. We exhibit an explicit order-3 fixed-point-free isometry s (trace -12), giving Λ the structure of a $\mathbb{Z}[\omega]$ -module with $\Lambda/\theta\Lambda \cong \mathbb{F}_3^{12}$ for $\theta = 1 - \omega$. Index-3 sublattices $M \supset \theta\Lambda$ are 3-elementary, so their theta series are weight-12 modular forms for $\Gamma_0(3)$. Decomposing Θ_M in $M_{12}(\Gamma_0(3)) = \langle E_{12}(\tau), E_{12}(3\tau), \Delta(\tau), \Delta(3\tau), g \rangle$, we find g present with *nonzero* coefficient. There are exactly two orbits of such sublattices under the complex-Leech automorphism group; both yield nonzero coefficients,

$$-\frac{3^7}{17 \cdot 73} \quad \text{and} \quad \frac{2^6 3^5}{17 \cdot 73}.$$

The common denominator $17 \cdot 73$ is exactly the pair of primes at which g is congruent to $\Delta \pmod{17}$ and to the Eisenstein series $\pmod{73}$. Every assertion is checked by direct computation, including a verification one coefficient *past* the Sturm bound.

1 Statement

Let $\omega = e^{2\pi i/3}$ and $\theta = 1 - \omega$, so $\mathbb{Z}[\omega]$ is the Eisenstein integers and $N(\theta) = 3$. Let Λ denote the Leech lattice (rank 24, even, unimodular, minimal norm 4). A *complex structure* on Λ is an order-3 isometry s with no nonzero fixed vectors; equivalently s satisfies $1 + s + s^2 = 0$, making Λ a free $\mathbb{Z}[\omega]$ -module of rank 12 with ω acting as s . This is the *complex Leech lattice*.

Let g be the unique normalized cuspidal newform in $S_{12}(\Gamma_0(3))$, with q -expansion $g = q + 78q^2 - 243q^3 + 4036q^4 - 5370q^5 + \dots$. The space $M_{12}(\Gamma_0(3))$ is 5-dimensional, with basis

$$E_{12}(\tau), \quad E_{12}(3\tau), \quad \Delta(\tau), \quad \Delta(3\tau), \quad g,$$

the first four spanning the old/Eisenstein part and g spanning the 1-dimensional new cuspidal part.

Theorem 1 (verified computationally). *Let $M \subset \Lambda$ be an index-3 sublattice containing $\theta\Lambda$. Then M is 3-elementary, its theta series $\Theta_M(\tau) = \sum_{x \in M} q^{|x|^2/2}$ lies in $M_{12}(\Gamma_0(3))$, and the coefficient of the newform g in Θ_M is nonzero. There are exactly two orbits of such M under $\text{Aut}(\Lambda, s)$; the g -coefficients are $-3^7/(17 \cdot 73)$ and $2^6 3^5/(17 \cdot 73)$ respectively.*

The remainder of this note records how each ingredient was constructed and checked.

2 The Leech lattice, certified

We build $\sqrt{8}\Lambda \subset \mathbb{Z}^{24}$ (integral coordinates, minimal norm 32) from the binary Golay code, then verify it three independent ways. The binary Golay code $[24, 12, 8]$ was constructed as the extended quadratic-residue cyclic code and confirmed to have weight enumerator $\{0:1, 8:759, 12:2576, 16:759, 24:1\}$ (in particular 759 octads). The lattice was generated by its three minimal-vector shapes $(\pm 4^2, 0^{22})$, $(\pm 2^8, 0^{16})$ on octads, and $(\mp 3, \pm 1^{23})$.

Certificate 1: kissing number. Direct enumeration of minimal vectors gives exactly 196560, and $\det = 8^{12} = 68719476736$, the defining invariants of $\sqrt{8}\Lambda$.

Certificate 2: automorphism group. The Plesken–Souvignier algorithm returns $\text{Aut}(\Lambda)$ of order

$$|\text{Co}_0| = 8,315,553,613,086,720,000,$$

Conway’s group $\text{Co}_0 = 2.\text{Co}_1$. No other 24-dimensional lattice has this automorphism group.

Certificate 3: the two-distance structure. Fixing a minimal vector v , the inner products $\langle u, v \rangle$ over all 196560 minimal vectors u are distributed as

$\langle u, v \rangle$	32	16	8	0	−8	−16	−32
#	1	4600	47104	93150	47104	4600	1

summing to 196560. This distribution is a fingerprint of the Leech lattice. In particular, every minimal vector has exactly **4600** minimal vectors at the minimal distance $|u - v|^2 = 64 - 2\langle u, v \rangle = 32$ (the value $\langle u, v \rangle = 16$).

Remark 1 (a recent extremality theorem). The quantity $4600/196560$ is precisely the statistic appearing in Bertucci and Bonifacio, *Bootstrapping Euclidean Lattices* (arXiv:2606.13771), where a conformal-bootstrap / semidefinite-programming bound on the mean-square number of minimal vectors at minimal distance from each minimal vector is shown to be saturated in dimensions 2, 4, 8, 24 by the hexagonal, D_4 , E_8 , and Leech lattices. Our direct computation of the value 4600 provides an independent check of the Leech entry. We emphasize that the spectrum entering that theorem is the *flat-torus* Laplace spectrum $\mathbb{R}^{24}/\Lambda$, whose eigenvalues are the dual-lattice norms—data equivalent to the theta series—and is unrelated to the Maass (hyperbolic) spectrum.

3 The order-3 isometry and the $\mathbb{Z}[\omega]$ -structure

An order-3 fixed-point-free isometry s was extracted from the generators of Co_0 . It is pinned down by a rigid constraint: from $1 + s + s^2 = 0$ one gets $\langle v, sv \rangle = -\frac{1}{2}|v|^2$ for every v , so on a minimal vector sv must be a minimal vector with $\langle v, sv \rangle = -16$ (one of the 4600). The element found satisfies all of:

$$s^3 = I, \quad \text{tr}(s) = -12, \quad s^\top G s = G, \quad \chi_s(x) = (x^2 + x + 1)^{12}, \quad \det(s - I) \neq 0,$$

and indeed $|v|^2 = |sv|^2 = 32$, $\langle v, sv \rangle = -16$ on minimal vectors. Thus $\omega := s$ makes Λ a rank-12 $\mathbb{Z}[\omega]$ -module: the complex Leech lattice, now explicit.

Writing $\theta = 1 - \omega = I - s$, we find

$$\det(I - s) = 3^{12}, \quad \Lambda/\theta\Lambda \cong (\mathbb{Z}/3)^{12} = \mathbb{F}_3^{12},$$

the Smith normal form of $(I - s)$ having elementary divisors $\{1^{12}, 3^{12}\}$. Moreover $|\theta v|^2 = 3|v|^2$ for all v , so $\theta\Lambda$ is a copy of $\sqrt{3}\Lambda$ and $\Theta_{\theta\Lambda}(\tau) = \Theta_{\Lambda}(3\tau)$. The quotient $\mathbb{F}_3^{12}$ is exactly the ambient space of the ternary Golay code—the Golay connection that resisted naive constructions lives in this quotient, not in a reduction map.

Remark 2 (why direct constructions failed). Building the complex Leech directly over $\mathbb{Z}[\omega]$ repeatedly produced the wrong lattice: the naive Construction A from the ternary Golay code has minimal norm 3 (the vectors θe_i), and the variant $\{x \bmod \theta \in C, \sum x_i \equiv 0 \bmod 3\}$ has kissing number 33264. Building the *real* Leech first—where 196560 is directly checkable—and then extracting s from Co_0 succeeds where reconstruction from memory does not.

4 Coset theta and the appearance of g

The whole-lattice theta Θ_{Λ} is a level-1 form, and the sublattice theta $\Theta_{\theta\Lambda} = \Theta_{\Lambda}(3\tau)$ has only 3-divisible exponents—both lie in the old/ V_3 image and carry no g . The new ingredient is the family of *index-3 sublattices* M with $\theta\Lambda \subseteq M \subset \Lambda$, corresponding to hyperplanes in $\Lambda/\theta\Lambda \cong \mathbb{F}_3^{12}$. Each such M has discriminant form $M^*/M \cong (\mathbb{Z}/3)^2$ (verified: Smith divisors $\{1^{22}, 3^2\}$), hence is 3-elementary, hence $\Theta_M \in M_{12}(\Gamma_0(3))$.

For an explicit M in the first orbit, direct enumeration to the Sturm bound (which is 4 for $M_{12}(\Gamma_0(3))$) gives

$$\Theta_M = 1 + 65340q^2 + 5593176q^3 + 132667794q^4 + \dots$$

(note the q^2 coefficient is not divisible by 3: this form is genuinely outside the oldform image). Solving the resulting 5×5 linear system over $\mathbb{Q}$ yields

$$\Theta_M = \frac{88573}{265720} E_{12}(\tau) + \frac{177147}{265720} E_{12}(3\tau) - \frac{350577}{11747} \Delta(\tau) + \frac{14348907}{11747} \Delta(3\tau) \boxed{-\frac{2187}{1241} g}.$$

The g -coefficient is $-2187/1241 = -3^7/(17 \cdot 73) \neq 0$.

Verification past the Sturm bound. The decomposition predicts the next coefficient $a_5 = 1543139208$. Independent enumeration of norm-10 vectors of M gives exactly 1543139208. Since this coefficient lies beyond the Sturm bound that guaranteed uniqueness, the agreement confirms both that Θ_M is precisely the stated form and that the g -coefficient is correct.

5 The two orbits

Hyperplanes in $\mathbb{F}_3^{12}$ are classified by an orbit invariant computable from minimal vectors alone: the number $a_2 = \#\{\text{minimal vectors of } \Lambda \text{ in } M\}$. A sweep finds exactly two values, hence two orbits. Running the full computation on a representative of each gives Table 1.

orbit	a_2	Θ_M low coefficients	g -coefficient	factored
1	65340	1, 0, 65340, 5593176, 132667794	$-2187/1241$	$-3^7/(17 \cdot 73)$
2	66798	1, 0, 66798, 5575680, 132746526	$15552/1241$	$2^6 3^5/(17 \cdot 73)$

Table 1: The two hyperplane orbits. Both give a nonzero g -coefficient with denominator $17 \cdot 73$.

Two structural features stand out. First, g is present in *both* orbits—it is never absent. Second, the Eisenstein part is *identical* across orbits,

$$\frac{88573}{265720} E_{12}(\tau) + \frac{177147}{265720} E_{12}(3\tau), \quad 177147 = 3^{11}, \quad 88573 = 265720 - 3^{11},$$

as Siegel–Weil demands: the Eisenstein part is the genus average, determined by local data common to both 3-elementary sublattices, while the cuspidal content (the Δ and g coefficients) is what distinguishes the orbits.

6 The role of 17 and 73

The denominator $1241 = 17 \cdot 73$ is not an artifact of the linear algebra; it is forced by the congruence structure of g . Direct computation of $a_p(g)$ for all primes $p < 40$, $p \neq 3$, confirms

$$a_p(g) \equiv \tau(p) \pmod{17}, \quad a_p(g) \equiv 1 + p^{11} \pmod{73},$$

that is,

$$\boxed{g \equiv \Delta \pmod{17}} \quad \text{and} \quad \boxed{g \equiv E_{12} \pmod{73}}.$$

The first is a congruence between the newform and the old cusp form Δ ; the second is an Eisenstein congruence between the newform and the boundary. To separate g from the old/Eisenstein part inside the integral lattice of forms one must invert 17 and 73, so the coefficient of (normalized) g in any integral theta decomposition has denominator divisible by $17 \cdot 73$. Both orbits realize exactly this: the numerators -3^7 and $2^6 3^5$ are coprime to $17 \cdot 73$, so the denominator is exactly the congruence number. The primes 17, 73 (together with the Ramanujan prime 691, visible in $11747 = 17 \cdot 691$ and the Eisenstein normalization $65520/691$) are the same old/new/Eisenstein primes that organize the level-3 weight tower.

7 Scope

The result is a statement about *holomorphic* modular forms: the complex Leech lattice sees the holomorphic newform $g \in S_{12}(\Gamma_0(3))$ through the theta series of its index-3 sublattices. It says nothing about the Maass (real-analytic) spectrum, about Ramanujan–Petersson bounds on Hecke eigenvalues (which remain Deligne’s theorem), or about the Riemann Hypothesis. Theta series of lattices are holomorphic forms and never produce Maass eigenvalues; the entire construction lives on the holomorphic side. The specific values $-3^7/(17 \cdot 73)$ and $2^6 3^5/(17 \cdot 73)$ depend on the chosen sublattice; the orbit-independent content is that g appears with nonzero coefficient, with denominator exactly the congruence primes $17 \cdot 73$.

Appendix: computational summary

All computations were performed in PARI/GP (via `cypari2`) and `numpy`.

- Binary Golay [24, 12, 8]: extended QR cyclic code; weight enumerator verified.
- $\sqrt{8}\Lambda$: generated from the three minimal-vector shapes; `qfminim` gives kissing 196560, $\det 8^{12}$.
- $\text{Aut}(\Lambda)$: `qfauto` returns order $|\text{Co}_0|$ in ≈ 14 s.
- Inner-product distribution: $\langle u, v \rangle$ over all minimal vectors for fixed v .
- s : an order-3, trace(-12) word in the Co_0 generators; all five defining identities checked; save/reload round-trip verified.
- $\theta = I - s$: $\det = 3^{12}$, Smith form $\{1^{12}, 3^{12}\}$, $|\theta v|^2 = 3|v|^2$.

- M : index-3 sublattice from a functional vanishing on $\theta\Lambda$; LLL-reduced; $M^*/M \cong (\mathbb{Z}/3)^2$.
- Θ_M coefficients: `qfminim` counts (no storage) to norm 8 (Sturm bound) and to norm 10 (post-Sturm check).
- Newform g and basis: PARI `mf` package (`mfinit`, `mfeigenbasis`, `mfcoefs`); Hecke relation $a_4 = a_2^2 - 2^{11} = 4036$ checked.
- Congruences: $a_p(g) \bmod 17$ vs $\tau(p)$, and $a_p(g) \bmod 73$ vs $1 + p^{11}$, for all primes $p < 40$.